

# Auto Sales Data Governance & Monthly Incremental Pipeline

Production data warehouse for Fortune-50 automaker | 5-month solo project | Jan-May 2026

| Pipeline Runtime | Tables Designed     | Wide Table Rows | Data Coverage    |
|------------------|---------------------|-----------------|------------------|
| 30 min/month     | 18                  | 71,304          | 99.92%           |
| (was 1-2 days)   | (8-layer warehouse) | (38 columns)    | (8/8 audit PASS) |

## PROBLEM & SOLUTION

Analyst team manually merged 3 Excel files monthly for sales reporting; brand/model naming inconsistencies caused silent errors with no audit trail. Designed and shipped a complete MySQL-based 8-layer data warehouse + monthly incremental pipeline that automated the entire workflow.

## KEY CONTRIBUTIONS

- **Architecture:** Designed 8-layer warehouse (RAW → STG → DIM → STD → DWD → DWS → AUDIT) with 18 tables; documented 25 architectural decisions (ADR-style) and 11 issues.
- **Algorithm:** Implemented Largest Remainder Method in pure MySQL window functions for integer volume allocation; guarantees 100% conservation across 1,939 buckets x 12 months.
- **Bug Discovery:** Found and fixed a critical 27% data loss bug (504,955 cars/month) in the wide table; restructured to bucket-union + sentinel-row pattern.
- **Business Workflow:** Co-designed 4-bucket new-model triage (APPROVE/MERGE/DEFER/REJECT) with business analysts; reduced analyst review time by 70%.
- **Code:** Wrote 1,700+ lines of production MySQL SQL (CTEs, window functions, idempotent UPSERTs); 3 Python adapters for Excel-to-CSV ingestion.
- **Documentation:** Authored 200+ KB handoff documentation (READMEs, SOPs, decision logs) enabling software engineering team to take over for productionization.

## TECH STACK

**Database:** MySQL 8.0.45 (utf8mb4\_0900\_ai\_ci) | **Code:** Python 3 (pandas, openpyxl) for ETL adapters | **Algorithms:** LRM apportionment, sentinel-row pattern, surrogate key strategy | **Quality:** 8-class automated audit framework (conservation, completeness, range, uniqueness)

## LIVE DEMO & PORTFOLIO

**Interactive demo:** <https://YOUR-APP-NAME.streamlit.app> — LRM algorithm playground, architecture explorer, data flow simulation, decision records.  
**Code:** [github.com/YOUR-USERNAME/auto-sales-pipeline](https://github.com/YOUR-USERNAME/auto-sales-pipeline) (public; anonymized data only).  
**Confidentiality:** Original project handled confidential business data; all numbers and names shown are anonymized for portfolio.